MILITARY SPECIFICATION

CABLES, SPECIAL PURPOSE, ELECTRICAL, NON-SHIELDED

(SHIPBOARD USE)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification describes the requirements for the procurement of one type of electric cable suitable for high temperature, high frequency use. The cable consists of an unshielded, unjacketed, twisted pair of conductors. The voltage rating of the cable is 600 volts root mean square (r.m.s.), and the temperature range is minus 73 to 200 degrees centigrade (C.).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation of bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

J-C-98	Cables and Wires, Insulated; Methods of Sampling and Testing
QQ-W-343	Wire, Electrical and Non-electrical, Copper (Uninsulated)

MILITARY

MIL-C-915	Cable, Cord and Wire, Electrical (Shipboard Use)

FSC 1290

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MIL-T-5438	Tester; Abrasion, Electrical Cable
MIL-C-12000	Cable, Cord, and Wire, Electric, Packaging and Packing of
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-W-21927	Weapons, Handling and Preparation for Delivery of; General Requirements for

STANDARDS

MILITARY

MIL-STD-104	Limits for Electrical Insulation Color

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN INSTITUTE FOR TESTING MATERIAL PUBLICATION

ASTM B 298	Silver-Coated Soft or Annealed Copper Wire

(Application for copies should be addressed to the American Institute for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

AMERICAN TRUCKING ASSOCIATION, INC., PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street, N.W., Washington 6, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

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(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Pre-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the control motor shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. Provision of samples shall be in accordance with 4.1.1.

3.1.2 Materials.

3.1.2.1 Quality. - All material used in the construction of completed cable shall be of the quality and form best suited for the purpose intended.

3.1.2.2 Cable components.

3.1.2.2.1 Conductors. - The copper strands of the conductors shall be soft or drawn-and-annealed copper strands, coated with commercially pure silver for a coating thickness not less than 40 microinches.

3.1.2.2.2 Conductor insulation. - The conductor insulation shall be of polytetrafluoroethylene.

3.1.3 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance and design of the contractor's inspection gages. However, such procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of the cable by Navy Final Inspection Gages.

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3.1.4 Requests for deviation. - Each request that a deviation be authorized shall have been submitted to the procuring activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the procuring activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.1.5 Conflicting requirements. - Conflicting requirements arising between this specification and any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.2 Definitions. - The definitions of terms in Specification MIL-C-915 shall be applicable to this specification.

3.3 Performance requirements and product characteristics. - The cable shall meet the following performance requirement and product characteristics:

3.3.1 Construction requirements.

3.3.1.1 Strands.

3.3.1.1.1 Diameter of strands. - When measured in accordance with 4.4.4.1.1, the diameter of the strands of each conductor shall be in accordance with table I with the tolerances of table II applicable. See 4.4.4.1.1.

Table II. Strand Diameter Tolerances

Nominal Diameter	Tolerances
Under 10 mils	+ 0.3 mil - 0.1 mil
10 mils and over	+ 3 percent - 1 percent

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Table I. Cable Dimensions

Cable Size (AWG)	Conductors							Finished Cable		
	No. of Copper Strands	Diameter of Strands (Nominal)	Cross Sectional Strands (Nominal)	Diameter Over Strands (Nominal)		Diameter Over Insulation (Max.)	Average Insulation Thickness (Min.)	Heat Resistance Test Load	Lay (Max.)	Diameter Over Insulation (Max.)	Conductor Resist. per 1000 Feet at 20°C (Max.)
				Outer Layer	Inner Layer						
		Inch	Circular Mils	Inch	Inch	Inch	Inch	Pound(s)	Inch(es)	Inch	Ohms
28	7	.0050	175	.015	-	.039	.008	0.5	1.00	.078	72.6
26	7	.0063	278	.019	-	.043	.008	1.0	1.00	.086	45.6
24	19	.0050	475	.025	.015	.049	.008	1.5	1.25	.096	26.5
22	19	.0063	754	.032	.019	.056	.008	2.5	1.25	.108	16.7
20	19	.0080	1216	.040	.024	.064	.008	4.0	1.50	.124	10.3
18	19	.0100	1900	.050	.030	.074	.008	6.0	1.75	.148	6.63
16	19	.0113	2426	.057	.034	.083	.009	8.0	2.00	.174	5.21
14	19	.0142	3831	.071	.043	.101	.009	12.0	2.50	.202	3.31

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3.3.1.2 Stranded conductors.

3.3.1.2.1 Type of stranding. - When checked in accordance with 4.4.4.2.1 each conductor shall be of concentric or unilay construction. See 4.4.4.2.1.

3.3.1.2.2 Number of strands. - When checked in accordance with 4.5.1.1.1, the number of copper strands in each conductor shall be as specified in table I. See 4.4.4.2.2.

3.3.1.2.3 Diameter. - The diameter over the strands of each stranded conductor shall be specified in table I within the following tolerances, as applicable. See 4.4.4.2.3.

AWG Size	Tolerances
26 and 28	+0.9 mil -0.3 mil
20, 22 and 24	+1.5 mil -0.5 mil
14, 16 and 18	+ 3 percent - 1 percent

3.3.1.2.4 Length of lay of strands. - When measured in accordance with 4.4.4.2.4 the length of lay of each layer of strands of each conductor shall not be less than 8 nor more than 16 times the nominal diameter of that layer as specified in table I. See 4.4.4.2.4.

3.3.1.2.5 Direction of lay of strands. - When checked in accordance with 4.4.4.2.5, the direction of lay shall be left-hand for the outer layer of strands of each conductor and shall be reversed for each successive inner layer. See 4.4.4.2.5.

3.3.1.3 Insulation.

3.3.1.3.1 Application of insulation. - When the insulation of each conductor is checked in accordance with 4.4.4.3.1, it shall consist of a tight fitting tube of uniform thickness of polytetrafluoroethylene which is extruded or taped on. If the insulation is taped on, the convolutions shall be fused to form a smooth homogenious tube, but a slight spiral ridge shall not be considered cause for rejection. See 4.4.4.3.1.

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3.3.1.3.2 Insulation thickness.

3.3.1.3.2.1 Average thickness. - When measured in accordance with 4.4.4.3.2.1, the average thickness of the insulation of each conductor shall meet the requirements of table I. See 4.4.4.3.2.1.

3.3.1.3.2.2 Minimum thickness. - When measured in accordance with 4.4.4.3.2.2, the minimum thickness of the insulation of each conductor shall not be less than 70 percent of the maximum thickness, both measurements being made at the same cross section. See 4.4.4.3.2.2.

3.3.1.3.3 Diameter over insulation. - When measured in accordance with 4.4.4.3.3, the diameter of each conductor over the insulation shall meet the requirements of table I. See 4.4.4.3.3.

3.3.1.3.4 Stripes.

3.3.1.3.4.1 Width. - The width of the wide stripe shall be not less than .031 inch when the diameter over the marked surface is greater than .047 inch, and not less than two thirds the diameter for diameters equal to or less than .047 inch. The width of the narrow stripe shall be approximately one half the width of a wide stripe and not less than 0.015 inch. See 4.4.4.3.4.

3.3.1.3.4.2 Spacing. - For single strip conductors, the spacing between stripes shall be not less than twice the width of the wide stripe. For multiple stripe conductors, the width of the space between two adjacent stripes shall be not less than the succeeding stripe width. See 4.4.4.3.4.

3.3.1.3.4.3 Lay. - The maximum length of lay shall be 1.00 inch. See 4.4.4.3.4.

3.3.1.3.5 Color. - The insulation colors shall comply with Standard MIL-STD-104, and shall not fail to comply under exposure to heat. See 4.4.4.3.5.

3.3.1.4 Cabling. - The cable shall consist of two uninsulated, stranded conductors twisted together, without fillers, with diameter over the insulation, and lay, as given in table I.

3.3.2 Electrical and mechanical requirements.

3.3.2.1 Conductor resistance. - When measured in accordance with 4.4.4.5, the resistance of each conductor of the cable shall meet the requirements of table I. See 4.4.4.5.

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3.3.2.2 Dielectric strength. - When tested in accordance with 4.4.4.6, the cable shall withstand an alternating-current (a-c) potential of 2000 volts root-mean-square (r.m.s.) having approximately sinusoidal waveform and frequency between 25 and 100 cycles per second (c.p.s.) for 60 seconds without arcing, damage, or breakdown. See 4.4.4.6.

3.3.2.3 Insulation resistance. - When measured in accordance with 4.4.4.7, the insulation resistance in megohms shall not be less than 1,000,000 divided by the length of the cable in feet (i.e. the insulation resistance expressed in megohms per 1000 feet shall not be less than 1000). See 4.4.4.7.

3.3.2.4 Heat resistance. - The cable shall withstand the heat resistance test of 4.4.4.8, and thereafter shall show no damage and shall meet the requirements of 3.3.2.2.

3.3.2.5 Cold bending. - The cable shall withstand the cold bending test of 4.4.4.9, and thereafter shall show no damage and shall meet the requirements of 3.3.2.2.

3.3.2.6 Abrasion resistance. - The cable shall withstand the abrasion resistance test of 4.4.4.10, i.e., the abrasion resistance of the conductor insulation shall be such that when 8 readings of inches of tape travel to failure are obtained in accordance with 4.4.4.10, the arithmetic average of the readings which are less than the arithmetic average of all 8 readings shall not be less than 17 inches. See 4.4.4.10.

3.3.2.7 Soldering resistance. - The cable shall withstand the soldering resistance test of 4.4.4.11, and thereafter shall show no damage and shall not have shrunk or elongated more than 1/16 inch. See 4.4.4.11.

3.3 Shock-transportation, storage, and handling. - The complete pack, after being subjected to the tests specified in 4.5.2, shall not rupture so as to spill contents or break seals on unit package.

3.4 Strand coating and tensile requirements.

3.4.1 Strands.

3.4.1.1 Silver coating.

3.4.1.1.1 Thickness. - Prior to stranding, the thickness of the silver coating shall be 40 microinches minimum when tested in accordance with method B of American Society for Testing Materials Standard ASTM-B-298. See 4.5.1.1.1.

3.4.1.1.2 Continuity. - Prior to stranding, the coating shall meet the requirements of the continuity test specified in 4.5.1.1.2. See 4.5.1.1.2.

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3.4.1.2 Tensile strength and elongation. - Prior to stranding, the tensile strength shall not exceed 40,000 pounds per square inch and the elongation shall be at least 10 percent for a 10-inch gage length when the tensile strength and elongation test procedure for soft or drawn-and-annealed solid wire and strands of Specification QQ-W-343 is followed. See 4.5.1.2.

3.5 Clean-up. - Prior to and after final assembly, all components and the cable shall be thoroughly cleaned of wire fragments and other foreign matter.

3.6 Workmanship. - The cable, including all components, shall be constructed and finished in a manner to assure compliance with the requirements of this specification. The workmanship shall be in accordance with high grade commercial practice covering this type of material.

3.6.1 The workmanship exhibited in the pre-production sample will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities acceptable to the Government. The Government preserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of three spools shall be manufactured by the methods and procedures proposed for the production lot. These samples shall be submitted to the Government inspector for pre-production tests at an activity designated by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

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4.1.2 Periodic production samples. - Randomly select from each production lot unit packages of cable in accordance with tables III and IV. Unless otherwise specified in the contract or purchase order, these units shall be submitted to a Bureau of Naval Weapons assigned activity to determine compliance of the samples with the requirements of the contract, specifications, and drawings. The unit packages of cable shall be considered satisfactory if they meet all specifications and pertinent drawing requirements and withstand the tests specified in 4.4. and 4.5. Periodic production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

Table III

Sampling for visual and dimensional examination and for strand coating and tensile tests

Number of Spools or Reels in Inspection Lot	Number of Spools or Reels in Sample	Acceptance Number (defectives)	Rejection Number (defectives)
15 or under	5	0	1
16 to 40	7	0	1
41 to 110	10	0	1
111 to 300	15	1	2
301 and over	25	1	2

Table IV

Sampling for environmental tests

Number of Spools or Reels in Inspection Lot	Number of Spools or Reels in Sample	Acceptance Number (defectives)	Rejection Number (defectives)
40 or under	2	0	1
41 to 110	3	0	1
111 to 300	4	0	1
301 and over	5	0	1

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4.2 Classification of tests. - The inspection and testing of the cable shall be classified as follows:

4.2.1 Acceptance tests. - These tests shall be accomplished on three spools from each lot submitted for acceptance under contract. These tests are detailed in 4.4.

4.2.2 Pre-production and periodic production tests. - Pre-production and periodic production tests shall be accomplished on samples selected as specified in 4.1.1 and 4.1.2 which are representative of the production of the cable after the award of the contract, to determine that the production meets the requirements of this specification. These tests are detailed in 4.5.

4.2.2.1 Action in case of failure.

4.2.2.1.1 Pre-production tests. - Acceptance shall be based on no failure to meet the tests of 4.5, and no other defects. Failure of the sample to comply with these requirements will result in rejection of the sample or the cessation of production, as determined by the procuring activity.

4.2.2.1.2 Periodic production tests. - Acceptance shall be based on no more failures to meet the tests of 4.5 than are permitted in tables III and IV, and no other defects. Failure to comply with these requirements shall result in rejection of the lot or the cessation of production, as determined by the procuring activity.

4.2.3 Tests after delivery. - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all the tests specified herein, as deemed necessary, after installation of the cable. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification.

4.3.1 Micrometer. - A micrometer having a minimum capacity of 0.5 inch, capable of reading 0.0001 inch, with an accuracy of plus or minus 0.0001.

4.3.2 Steel scale. - A flexible steel scale graduated in thirty-seconds of an inch, minimum length - six inches.

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4.3.3 Color limit cards. - Supplement to Standard MIL-STD-104.

4.3.4 Pocket comparator. - A 10-power magnification pocket comparator.

4.3.5 Wheatstone bridge. - A direct-current resistance bridge capable of measuring resistance from 1 to 10,000 ohms with an accuracy of plus or minus 0.5 percent.

4.3.6 High potential test tank. - A tank containing a 0.5 to 1.0 percent water solution of a wetting agent such as Aerosol at room temperature in which 20 feet of cable may be immersed. The tank shall have a grounding electrode. The grounding electrode may be the surface of the tank if the surface is metal and in contact with the water.

4.3.7 High potential tester. - A high potential tester capable of applying up to 2000 volts r.m.s., 60 c.p.s., and having a volt-ampere rating of 1000 minimum. When the unit is turned on or off, the output voltage shall change uniformly with time, and shall reach 100 percent or zero percent rated output in not less than 10 nor more than 60 seconds.

4.3.8 Insulation resistance tester. - An insulation tester capable of measuring resistance to 100,000 megohms, using a direct-current test voltage not less than 100 nor greater than 500 volts.

4.3.9 Heat resistance test equipment. - An air oven capable of maintaining a temperature between 245 and 255 degrees C for 96 hours, equipped to perform the heat resistance test of 4.4.4.8.

4.3.10 Cold bending test equipment. - A cold chamber capable of maintaining a temperature of minus 73 degrees centigrade for a period of 4 hours, equipped to perform the cold bending test of 4.4.4.9.

4.3.11 Abrasion resistance tester. - An abrasion resistance tester which complies with the requirements of Specification MIL-T-5438 with the following exceptions:

(a) Abradant. 1-inch by 50 yard abradant tape 400-grit aluminum oxide.
(b) Fault-signal device. Sensitivity of 1 milliampere or less.
(c) Conducting strip. Conducting silver paint marked every inch.

4.3.12 Solder resistance test equipment. - Pot of 60 percent tin, 40 percent lead solder maintained at approximately 320 degrees centigrade, set of mandrels with mandrel diameter equal to diameter over insulation of each size of stranded, insulated conductor, and timer capable of indicating intervals of 5 seconds with a minimum accuracy of plus or minus 0.1 second.

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4.4 Acceptance tests.

4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to assure that the cable is in compliance with the requirements of 3.1, 3.3.2.1, 3.3.2.2, 3.5, and 3.6. Manufacturer's certification of materials specified in the applicable drawings shall be used in lieu of material tests.

4.4.2 Test conditions. - Unless otherwise specified, the cable shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature: Room ambient 65 to 95 degrees Fahrenheit (18 to 35 degrees centigrade).

4.4.2.2 Altitude: Normal ground atmospheric pressure.

4.4.2.3 Vibration: None.

4.4.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure. - The detailed test procedure listed in the following paragraphs shall be followed.

4.4.4.1 Strands.

4.4.4.1.1 Diameter. - The diameter of 7 strands of each stranded conductor shall be measured using the micrometer of 4.3.1 to determine compliance with the requirements of 3.3.1.1.1. The procedure given in Specification QQ-W-343 for examination of solid wire and strand diameters shall be followed.

4.4.4.2 Stranded conductors.

4.4.4.2.1 Type of stranding. - The insulation shall be removed from several inches of conductor, and the stranding shall be inspected visually to determine compliance with the requirements of 3.3.1.2.1.

4.4.4.2.2 Number of strands. - The number of copper strands in each conductor shall be counted to determine compliance with the requirements of 3.3.1.2.2.

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4.4.4.2.3 Diameter. - The diameter over each stranded conductor shall be measured using the micrometer of 4.3.1 to determine compliance with the requirements of 3.3.1.2.3. The procedure given in Specification QQ-W-343 for examination of solid wire and strand diameters shall be followed.

4.4.4.2.4 Length of lay of strands. - The length of lay of each layer of strands of each conductor shall be measured using the steel scale of 4.3.2 to determine compliance with the requirements of 3.3.1.2.4.

4.4.4.2.5 Direction of lay of strands. - The direction of lay of the outer layer of strands of each conductor shall be observed to determine compliance with the requirements of 3.3.1.2.5.

4.4.4.3 Insulation.

4.4.4.3.1 Application of insulation. - The insulation of each conductor shall be checked visually to determine compliance with the requirements of 3.3.1.3.1.

4.4.4.3.2 Insulation thickness.

4.4.4.3.2.1 Average thickness. - The average thickness of the insulation of each conductor shall be measured using method 1011, 1013, or 1018 of Specification J-C-98 to determine compliance with the requirements of 3.3.1.3.2.1.

4.4.4.3.2.2 Minimum thickness. - The minimum and maximum thickness of the insulation of each conductor shall be measured at the same cross section, and the ratio of minimum thickness to maximum thickness shall be computed to determine compliance with the requirements of 3.3.1.3.2.2. The minimum thickness of insulation shall be measured in accordance with method 1011, 1013, or 1018 of Specification J-C-98.

If method 1011 or 1013 of Specification J-C-98 is used, the diameter over the insulated conductor at the point where the insulation is considered thinnest shall be measured before removing the insulation on the side opposite this point so that the maximum insulation thickness at the cross section can also be computed, i.e., the maximum insulation thickness is the diameter over the insulated conductor at the point considered thinnest minus the same measurement made with the insulation removed from the side of the conductor opposite this point.

If method 1018 is used, the maximum insulation thickness shall be determined using the same procedure specified for determining the minimum insulation thickness.

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4.4.4.3.3 Diameter over insulation. - The micrometer of 4.3.1 shall be set to the maximum diameter over insulation as given in table I, and approximately two feet of the insulated conductor shall be fed through the micrometer to determine compliance with the requirements of 3.3.1.3.3. The insulated conductor shall be considered in compliance with the requirements of 3.3.1.3.3 if it passes through the micrometer without sticking.

4.4.4.3.4 Stripes. - The width, spacing, and lay of the stripes (if stripes are present) shall be measured using the scale of 4.3.2 and the pocket comparator of 4.3.4, as applicable, to determine compliance with the requirements of 3.3.1.3.4.

4.4.4.3.5 Colors. - The insulation colors shall be checked visually against the color limit cards of the Supplement to Standard MIL-STD-104 to determine compliance with the requirements of 3.3.1.3.5.

4.4.4.4 Cabling. - The cable shall be inspected visually, the cable diameter shall be checked using the micrometer of 4.3.1, and the lay of the cable shall be measured using the scale of 4.3.2 to determine compliance with the requirements of 3.3.1.4. The diameter shall be considered in compliance with the requirements of 3.3.1.4 if no sticking occurs when approximately two feet of cable are passed through the micrometer of 4.3.1 with the micrometer set to the maximum cable diameter specified in table I.

4.4.4.5 Conductor resistance. - The resistance of the entire length of cable on the spool (each conductor separately) shall then be measured using the Wheatstone bridge of 4.3.5, and the resistance of the cable shall be divided by the length of the cable in feet as marked on the spool or reel to determine compliance with the requirement of 3.3.2.1.

4.4.4.6 Dielectric strength. - The entire spool length of cable shall be immersed for a minimum of 4 hours in the high potential test tank of 4.3.6. At the end of 4 hours the high potential voltage shall be applied for 60 seconds between conductors and between each conductor and the other grounded using the high potential tester of 4.3.7 to determine compliance with the requirements of 3.3.2.2.

4.4.4.7 Insulation resistance. - Immediately following the dielectric strength test of 4.4.4.6, and with the cable still immersed, the insulation resistance between each conductor and the other conductor grounded shall be measured using the insulation resistance tester of 4.3.8 to determine compliance with the requirements of 3.3.2.3. The length of the cable shall be taken as the length marked on the spool or the reel.

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4.4.4.8 Heat resistance. - The cable shall be untwisted and a 3 foot specimen of each conductor shall be wound 5 close turns around a mandrel having a diameter of equal to or slightly greater than 3 times the insulated conductor diameter. The mandrel shall be mounted horizontally in the oven of 4.3.9 with each end of the conductor loaded with the weight specified in table I. The oven shall be maintained at a temperature of 245 to 255 degrees centigrade (473 to 491 degrees Fahrenheit) for 96 hours. The mandrel and weight loaded conductors shall then be removed from the oven and allowed to cool to room temperature. One hour after the mandrel has reached room temperature, the weights shall be removed and the conductors shall be removed from the mandrel and straightened. The conductors shall then be subjected to the procedure of 4.4.4.6 (with the specimen length required of 4.4.4.6 waived) to determine compliance with the requirements of 3.3.2.4 and to the procedure of 4.4.4.3.4 to determine compliance with the requirements of 3.3.1.3.5.

4.4.4.9 Cold bending. - A cable specimen having a length of 12 inches greater than the length required for winding on the mandrel as specified in table V shall be used for this test. The specimen shall be secured to the mandrel in such a manner that bending will occur at a distance at least 6 inches from one end of the specimen. A minimum of 2 turns shall be made on the mandrel at room temperature. The free end of the specimen shall be secured to the load weight specified in table V. The entire coiled specimen shall be placed in a cold chamber at a temperature of minus 73 degrees centigrade (minus 100 degrees Fahrenheit) for a period of 4 hours. While in the cold chamber at this temperature, the mandrel shall be rotated at an approximate rate of one turn per 4 seconds until a total of 6 turns of wire have been wound on the mandrel. The 6 turns of wire shall then be unwound from the mandrel and rewound on the mandrel in a direction opposite to the direction of the first winding. The coiled specimen shall be removed from the cold chamber after a period of 4 hours and brought to room temperature. No readily visible derangements shall result from the cooling or bending. When the cable specimen reaches room temperature, the dielectric strength of the coiled cable specimen shall be checked in accordance with 4.4.4.2.1 to determine its compliance with the requirements of 3.4.1.

Table V. Cold Bend Test Conditions

AWG Size	28	26	24	22	20	18	16	14
Mandrel Diameter (Inches)	3/8	7/16	1/2	9/16	5/8	3/4	7/8	1
Test Load (Pounds)	0.8	1.0	1.6	3.0	6.0	8.0	10.0	16.0

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4.4.4.10 Abrasion resistance. - This test shall be performed on each conductor using an abrasion resistance tester which complies with the requirements of Specification MIL-T-5438, with the following exceptions:

(a) Abradant. 1-inch by 50-yard abradant tape, 400-grit aluminum oxide,
(b) Fault detector sensitivity. 1 milliampere minimum,
(c) Conducting strip. Conducting silver paint marked every inch.

A 36-inch specimen of insulated conductor shall be removed from the cable, and shall be drawn between the thumb and finger of one hand by the other hand to straighten it. One inch of insulation shall then be removed from one end of the conductor specimen, and the conductor specimen shall be clamped in the abrasion resistance tester at this end. Prior to clamping the other end, the tension specified in table VI shall be applied to the conductor specimen to make it taut, and it shall be positioned (with the weight support bracket removed) so that it is tangent to the abradant tape.

Using the weight and weight support bracket type specified in table VI, the abradant tape shall be run until failure of the insulation (establishment of conduction) occurs, and the number of inches of tape travel to failure shall be observed and recorded. Seven additional readings of inches of tape travel to failure shall be taken, advancing the specimen 2 inches and rotating it 90 degrees before taking each additional reading. The arithmetic average of the eight readings, and the arithmetic average of the readings which are less than the arithmetic average of all eight readings shall be computed to determine compliance with the requirements of 3.3.2.6.

Table VI. Abrasion Resistance Test Conditions

Test Parameter	Cable Size (AWG)								
	32	30	28	26	24	22	20	18	16
Wire Tension Pounds	0.05	0.10	0.15	0.25	0.40	0.60	1.0	1.5	2.0
Weight Support Bracket Type	A	A	A	A	A	A	A	B	B
Weight to be Added - Pounds	0.25	0.25	0.25	0.50	0.50	0.50	0.50	0.75	0.75

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4.4.4.11 Soldering resistance. - The cable shall be subjected to the soldering test procedure of Specification MIL-W-16878 to determine compliance with the requirements of 3.3.2.7.

4.4.4.12 Preparation for delivery. - Preparation for delivery of the cables shall conform to section 5 of this specification.

4.5 Pre-production and periodic production sample tests.

4.5.1 Place of test. - Pre-production and periodic production sample tests shall be made at the place of manufacture and shall be witnessed by the Government inspector.

4.5.1.1 Strand coating and tensile requirements.

4.5.1.1.1 Thickness of coating. - The thickness of the silver coating of the copper strands of the conductors shall be measured in accordance with ASTM B298 to determine compliance with the requirements of 3.4.1.1.1.

4.5.1.1.2 Continuity of coating. - The continuity of the silver coating of the copper strands of the conductors shall be checked in accordance with ASTM B298 to determine compliance with the requirements of 3.4.1.1.2.

4.5.1.2 Tensile strength and elongation. - The tensile strength and elongation of the strands of a 10-inch gage length shall be measured using the tensile strength and elongation test procedure for soft drawn- and annealed solid wire and strands of Specification QQ-W-343 to determine compliance with the requirements of 3.4.1.2.

4.5.2 Shock-transportation, storage, and handling. - The complete pack, prepared as specified in section 5, shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927.

4.6 Waiver of tests. - Tests specified in 4.5.2 may be waived by the procuring agency for level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Functional tests. - At the conclusion of the tests specified in 4.5.2, leakage and other tests, as applicable, shall be conducted on the unit packages to determine freedom from rupture of seal and compliance with all the requirements of section 3.

4.8 Failure of the cable to meet any of the requirements and tests of this specification shall be considered cause for rejection.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preservative application. - Not applicable.

5.1.1.2 Unit packaging. - The cable shall be furnished on spools or reels in compliance with the level A packaging requirements of Specification MIL-C-12000, with both ends of the cable accessible for testing. In any "inspection lot", at least 80 percent of the total footage shall consist of lengths not less than 50 feet long, and none of the lengths of the remaining total footage shall be less than 10 feet long.

5.1.1.3 Intermediate package. - Not applicable.

5.1.1.4 Cushioning. - The unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices, to ensure that the cable will meet all the requirements of section 3 after being tested in accordance with 4.5.2.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. - Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, drying and preservative application. - Not applicable.

5.1.3.2 Unit packaging. - The cable shall be furnished on spools or reels in accordance with good commercial practice, with both ends of the cable accessible for testing, and sealed so as to be moisture proof.

5.1.3.3 Intermediate package. - Not applicable.

5.2 Packing.

5.2.1 Number of packages per pack. - The number of unit packages per pack shall be specified in the contract or purchase order.

5.2.2 Level A.

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5.2.2.1 Exterior containers. - Cables packaged as described in 5.1.1.2 shall be packed in compliance with the level A packing requirements of Specification MIL-C-12000. The type, grade, style, and size of the box shall be designated by the procuring activity.

5.2.3 Level B.

5.2.3.1 Exterior containers. - Cables packaged as described in 5.1.1.2 shall be packed in compliance with the level B packing requirements of Specification MIL-C-12000. The type, grade, style, and size of the box shall be designated by the procuring activity.

5.2.4 Level C. - Cables packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special markings. - None, unless otherwise specified.

5.3.2 Normal markings. - In addition to the normal markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Specification MIL-C-12000, including serial numbers or manufacturing lot numbers, and shall give the following:

(a) Name of manufacturer
(b) Date of manufacture
(c) Number and date of this specification
(d) Number of conductors
(e) AWG size
(f) Actual length of cable

6. NOTES

6.1 Intended use. - The cable is intended for use in the Angle Solvers Mark 18 Mod 2 and 3, the Fire Control System Mark 112 Mod 1, the Sonar Receiving Sets AN/BQG-1, -2, -3, -4, or other suitable application.

6.2 Ordering data. - Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specification; applicable drawings and other documents; selection of applicable levels of packaging and packing; number of unit containers per pack; guarantee clause (see 6.4); and total footage of each AWG size. In addition the color or color combinations required shall be specified.

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6.2.2 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-C-12000 and to par. 4.6.

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For level B. - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Guarantee. - The contract or order shall include a requirement for a guarantee stating that any cable which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

6.5 The contract or order shall include a sufficient number of extra items to cover production and pre-production testing samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented inventions that may in any way be related thereto.

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